Variation and selection

Mark scheme and worked solutions for 30 questions: 15 multiple choice, 15 structured.

1 · Part 1

Core Variation [1]

Answer: B

Explanation

Variation describes differences between individuals of the same species. A compares two different species, C compares two types of cell, and D describes the growth of one individual over time, not differences between individuals.

2 · Part 1

Core Variation [1]

Answer: D

Explanation

Body mass can take values throughout a range. The other examples have distinct categories without intermediate types.

3 · Part 1

Core Variation [1]

Answer: A

Explanation

Discontinuous variation produces separate categories. Measurement precision and a continuous range do not define it.

4 · Part 1

Core Variation [1]

Answer: C

Explanation

Identical genotypes differ in height only because of the nutrient supply, so the environment contributes to this continuous variation. A and B are not supported by the observation, and D is wrong because height is continuous, not discontinuous, variation.

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5 · Part 1 Core Variation [1]

Answer: D

Explanation

Mutation is the only way a new allele is formed. Meiosis and random fertilisation shuffle existing alleles into new combinations, and selective breeding only chooses among alleles that already exist.

6 · Part 1 Core Variation [1]

Answer: B

Explanation

Ionising radiation damages DNA and increases the rate of mutation. Temperature, oxygen concentration and diet are not recognised mutagens.

7 · Part 1 Supplement Variation [1]

Answer: C

Explanation

A gene mutation is a random change in the base sequence of DNA. Meiosis, fertilisation and crossing by a breeder recombine existing alleles but do not change the base sequence of a gene.

8 · Part 1 Supplement Variation [1]

Answer: A

Explanation

Mutation creates new alleles, and meiosis and random fertilisation produce new combinations of alleles, so all three are sources of genetic variation. Diet, exercise, nutrition, climate, temperature and light intensity are environmental influences, not genetic ones.

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9 · Part 1 Core Adaptive features [1]

Answer: B

Explanation

An adaptive feature is inherited and helps survival and reproduction in the relevant environment. Injuries and learned behaviour are not inherited, and an inherited feature is not adaptive unless it helps the organism survive and reproduce.

10 · Part 1 Core Adaptive features [1]

Answer: C

Explanation

Being active at night avoids the hottest part of the day, so the animal is less likely to overheat and loses less water by evaporation. The behaviour does not remove the need to drink, remove all competition or change how much energy a meal provides.

11 · Part 1 Supplement Adaptive features [1]

Answer: D

Explanation

A thick waxy cuticle reduces evaporation from the leaf surface. A thin cuticle, many stomata on the sunlit upper surface and large flat leaves would all increase water loss.

12 · Part 1 Core Selection [1]

Answer: A

Explanation

Individuals that survive and reproduce more successfully pass more of their alleles to the next generation. Need does not create a useful allele, large size is not always an advantage, and selection by humans is artificial selection, not natural selection.

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13 · Part 1 Core Selection [1]

Answer: C

Explanation

Selective breeding selects parents and offspring with desirable features over generations. Mating without selection does not improve the breed, hormone treatment does not change inheritance, and inserting a gene is genetic modification, a different process.

14 · Part 1 Supplement Selection [1]

Answer: A

Explanation

The antibiotic kills susceptible bacteria; the pre-existing resistant variants survive and reproduce, so resistance becomes more common. Bacteria cannot choose to become resistant, the antibiotic does not make every bacterium mutate, and susceptible bacteria survive less well, not better.

15 · Part 1 Supplement Selection [1]

Answer: B

Explanation

Artificial selection is directed by human choice of which individuals breed. Both forms produce inherited change, both act on inherited variation (not acquired features) and both depend on variation in the population.

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16 · Part 1 Core Variation [4]

Mark scheme

(a) **[2]**

- The volunteers are individuals of the same species with different hand spans. (1)
- Hand span is continuous variation because it spans a range of values. (1)

ALLOW a range of values between two extremes / no distinct groups

(b) **[2]**

- Inherited genetic differences. (1)
- Environmental influences such as nutrition during growth. (1)

ALLOW any named environmental factor, e.g. diet, exercise

Worked solution

Variation means differences between individuals of the same species, and the volunteers all belong to one species yet have different hand spans. Hand span can take any value between the smallest and largest measured, with no separate categories, so it is continuous variation. Continuous features like this are controlled by genes and are also affected by the environment, for example nutrition while the person was growing.

17 · Part 1 Core Variation [4]

Mark scheme

(a) **[2]**

- The phenotypes fall into a limited number of separate categories. (1)
- There are no intermediate ABO groups between those categories. (1)

ALLOW only four possible phenotypes

(b) **[2]**

- Discontinuous variation is usually caused by genes alone. (1)
- Continuous variation is influenced by both genes and environment. (1)

ALLOW the environment has little or no effect on discontinuous variation

Worked solution

Everyone surveyed fits into one of just four groups: A, B, AB or O. There is no blood group 'between' A and B, so the data form a limited number of separate categories with no intermediates, which is discontinuous variation. Discontinuous features are usually controlled by genes alone (here, the ABO alleles) and the environment does not change them. Continuous features, such as height or mass, are controlled by genes and also influenced by the environment, which is why they produce a range of values rather than categories.

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18 · Part 1 Core Variation [4]

Mark scheme

(a) **[2]**

- The range is 61 − 18 = 43 mm. (1)

ALLOW range stated as 18 mm to 61 mm

- Leaf length shows continuous variation. (1)

(b) **[2]**

- Measure many leaves, the same number from each plant, using the same measuring method. (1)
- Sample leaves at a similar age or position on each plant. (1)

ALLOW repeat and calculate a mean; measure to the same precision; plants grown in the same conditions

Worked solution

The range is the difference between the largest and smallest values: 61 − 18 = 43 mm. Leaf length can take any value between these extremes, so it is an example of continuous variation. To compare leaf length between two plants fairly, measure the same number of leaves from each plant, take leaves of a similar age or position on the plant, use the same ruler and measure to the same precision, and calculate a mean so that one unusually large or small leaf does not distort the comparison.

19 · Part 1 Core Variation [4]

Mark scheme

(a) **[2]**

- Mutation is genetic change. (1)

ALLOW a change in a gene / in the DNA / in a chromosome

- It can produce a new alternative form of a gene, an allele. (1)

(b) **[2]**

- Ionising radiation. (1)
- Some chemicals. (1)

ALLOW named examples, e.g. X-rays, gamma rays, ultraviolet radiation; chemicals in tobacco smoke / tar

Worked solution

A mutation is a genetic change, a change in the DNA of a gene or in a chromosome. Because the alleles of a gene are different versions of its base sequence, a change in that sequence creates a new version, so mutation is the way in which new alleles are formed. Mutations happen at random, but their rate is increased by ionising radiation such as X-rays, gamma rays and ultraviolet light, and by some chemicals such as those in tobacco smoke.

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20 · Part 1 Supplement Variation [4]

Mark scheme

(a) **[2]**

- It is a random change. (1)
- The change is in the base sequence of DNA. (1)

ALLOW a change in one or more bases / nucleotides

(b) **[2]**

- Meiosis produces genetically different gametes. (1)
- Random fertilisation combines gametes in different combinations. (1)

ALLOW random mating

Worked solution

A gene mutation is a random change in the base sequence of DNA in a gene; it may change the protein the gene codes for and so produce a new allele. Siblings differ even without new mutations because, during meiosis, each gamete receives a different combination of the parent's alleles, and at fertilisation it is random which sperm fuses with which egg. Random mating (which individuals breed together) also adds to variation in a population.

21 · Part 1 Core Variation [4]

Mark scheme

(a) **[2]**

- The seedlings began with the same genotype. (1)
- Different environments can affect a continuously varying feature such as mass. (1)

ALLOW only the environment (light intensity) differed, so it must be the cause

(b) **[2]**

- Keep water and mineral supply the same. (1)
- Compare plants of the same age under the same temperature. (1)

ALLOW any two of same water, minerals, temperature, carbon dioxide, soil, species, age, number of seedlings, time period

Worked solution

The seedlings are genetically identical, so their genes cannot be the cause of the difference. The only factor that differed was light intensity, so the difference in mass must be an effect of the environment. Mass is a continuous feature, which is influenced by both genes and the environment, so a difference in mass never by itself proves a difference in genes. To test light alone, every other factor that affects growth must be kept the same: water and mineral supply, temperature, carbon dioxide, soil, and the age and number of seedlings.

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22 · Part 1 Core Variation [4]

Mark scheme

(a) **[2]**

- Body length can have intermediate values across a range. (1)
- Seed shape has separate categories without intermediates. (1)

(b) **[2]**

- Measure lengths with a ruler using consistent units and record in a table with the unit in the heading. (1)
- Count (tally) the number of seeds in the smooth and wrinkled categories. (1)

ALLOW group lengths into size classes / plot a histogram; plot seed shape as a bar chart

Worked solution

Body length is continuous variation: any length between the smallest and largest is possible, so the data are a range of values. Seed shape is discontinuous: every seed is either smooth or wrinkled, with nothing in between, so the data are two categories. Continuous data are measured, for example with a ruler to the nearest millimetre, recorded in a table with the unit in the heading, and grouped into size classes for a histogram. Discontinuous data are counted: tally how many seeds are in each category and show the totals in a bar chart.

23 · Part 1 Supplement Variation [4]

Mark scheme

(a) **[2]**

- A gene mutation can create it. (1)

ALLOW mutation, unqualified, for this mark

- A gene mutation is a random change in the DNA base sequence. (1)

(b) **[2]**

- Meiosis produces gametes with different genetic combinations. (1)
- Random mating or random fertilisation brings together different existing alleles. (1)

ALLOW independent assortment / crossing over as detail of meiosis

Worked solution

Only mutation can create a base sequence that did not exist before: a gene mutation is a random change in the base sequence of DNA, which can give a new allele. Sexual reproduction cannot make new alleles, but it shuffles existing ones. Meiosis gives each gamete a different combination of the parent's alleles, random mating decides which two individuals breed, and random fertilisation decides which gametes fuse. Together these produce offspring with new combinations of alleles, increasing variation in the population.

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24 · Part 1 Supplement Adaptive features [4]

Mark scheme

(a) **[2]**

- Rolled leaves trap moist air and reduce air movement near stomata. (1)
- Sunken stomata retain humid air, reducing the water-vapour concentration gradient. (1)

> **ALLOW** reduces the water potential gradient / diffusion gradient; air next to stomata becomes saturated

(b) **[2]**

- A thick cuticle reduces evaporation from the leaf surface. (1)
- Storage tissue retains water for use when water uptake is limited. (1)

> **ALLOW** cuticle is waxy / waterproof; storage allows survival between rainfalls

Worked solution

Transpiration is the loss of water vapour from the leaf, mainly through the stomata, and it is faster when the air outside is dry and moving. Rolled leaves enclose the stomata so that still, humid air is trapped inside the roll, and sunken pits do the same on a smaller scale: the air in the pit becomes saturated with water vapour, so the concentration gradient between the inside of the leaf and the air just outside the stomata is small and diffusion is slow. A thick waxy cuticle is waterproof, so very little water evaporates through the leaf surface itself. Water-storage tissue holds water taken up during rare rainfall so the plant can keep its cells turgid and keep photosynthesising during long dry periods.

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25 · Part 1 Core Adaptive features [4]

Mark scheme

(a) **[2]**

- Webbed feet provide a larger surface for propulsion through water. (1)
- Waterproof feathers help keep the body dry and insulated. (1)

ALLOW swimming to find food / escape predators; feathers aid buoyancy / keep the bird warm

(b) **[2]**

- It must be inherited. (1)

ALLOW genetic / passed on from parents

- It must help survival and reproduction in that environment. (1)

Worked solution

Webbed feet spread out to give a large surface area that pushes against the water, so the bird swims efficiently to find food and escape predators. Waterproof feathers stop water reaching the skin, so the bird stays dry, keeps its body heat (insulation) and floats more easily. For a feature to count as an adaptive feature it must be inherited, that is, passed from parents to offspring through alleles, and it must help the organism to survive and reproduce in its particular environment.

26 · Part 1 Core Selection [4]

Mark scheme

(a) **[2]**

- Predators are less likely to detect and eat dark beetles. (1)

ALLOW camouflage; light beetles are eaten more

- More dark beetles survive to reproduce. (1)

(b) **[2]**

- Surviving dark beetles pass their alleles to offspring. (1)
- The alleles producing the advantageous colour become more common. (1)

ALLOW proportion of dark beetles rises each generation; natural selection

Worked solution

On dark ground, dark beetles are camouflaged, so predators find and eat fewer of them than light beetles. More dark beetles therefore survive long enough to reproduce. Because colour is inherited, the surviving dark beetles pass on the allele for dark colour to their offspring, while fewer light beetles breed. In each generation the proportion of dark beetles rises, so the frequency of the dark form in the population increases. This is natural selection.

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27 · Part 1 Core Selection [4]

Mark scheme

(a) **[2]**

- Select parents with high milk yield or a family history of high yield. (1)

ALLOW select males whose mothers or sisters have high milk yield

- Cross the selected individuals. (1)

(b) **[2]**

- Select offspring that show high milk yield. (1)
- Breed the selected offspring and repeat over many generations. (1)

ALLOW cross selected offspring with each other or with other high-yielding goats

Worked solution

Selective breeding starts by choosing parents that show the desired feature: the breeder selects female goats with the highest milk yield (and males whose mothers or sisters are high yielders, since males do not produce milk). These selected goats are crossed to produce the next generation. The offspring are then assessed, and only those with the highest milk yield are kept and bred together. Repeating this selection and crossing over many generations gradually increases the average milk yield of the herd.

28 · Part 1 Supplement Selection [4]

Mark scheme

(a) **[2]**

- Susceptible bacteria are killed or their reproduction is inhibited. (1)
- Resistant bacteria survive more successfully. (1)

ALLOW population size falls sharply

(b) **[2]**

- Surviving resistant bacteria reproduce. (1)
- Their offspring inherit resistance, increasing its frequency in the population. (1)

ALLOW less competition for resources; natural selection; a resistant strain develops

Worked solution

Antibiotics kill bacteria or stop them reproducing, so when treatment starts most of the population, the susceptible bacteria, die and the population falls sharply. The few bacteria that already carried a resistance allele (from a random mutation) are not killed. After treatment these survivors reproduce rapidly, with little competition from the susceptible bacteria that have died, and pass the resistance allele to their offspring. The proportion of resistant bacteria therefore increases until the population is mostly resistant. The antibiotic did not create the resistance; it selected for bacteria that already had it, which is natural selection.

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29 · Part 1 Supplement Selection [4]

Mark scheme

(a) **[2]**

- Inherited drought-tolerant variants survive and reproduce more successfully. (1)

ALLOW there is variation in drought tolerance; less tolerant plants die

- Their alleles become more common over many generations. (1)

(b) **[2]**

- Environmental conditions determine which wild variants reproduce most successfully. (1)
- The farmer chooses which cultivated plants reproduce for seed yield. (1)

ALLOW natural selection in the wild population, artificial selection in the farmed population

Worked solution

In the wild population there is genetic variation: some grass plants tolerate drought better than others. During a drought the less tolerant plants die, while the tolerant ones survive and reproduce, passing on their alleles. Over many generations the alleles for drought tolerance become more common and the population becomes more suited to its environment; this process is adaptation by natural selection. In the farmer's field, survival in the wild no longer decides reproductive success. The farmer chooses which plants provide seed for the next generation on the basis of seed yield, so it is human choice, artificial selection, that determines which alleles are passed on.

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30 · Part 1 Core Selection [4]

Mark scheme

(a) **[2]**

- Food and other resources are limited. (1)
- Many young compete for those resources and some die. (1)

ALLOW named resources (food, oxygen, space); predation

(b) **[2]**

- Individuals already differ in inherited features (variation arising from mutation, meiosis and random fertilisation). (1)
- Those with advantageous features survive and leave more offspring; an individual cannot choose or change its alleles. (1)

ALLOW the environment determines which variants survive

Worked solution

A pond has limited food, oxygen and space, so if many more young fish are produced than the pond can support, they must compete for these resources and many die: a struggle for survival. Which fish survive depends on inherited differences that already exist in the population, such as faster swimming or better camouflage, which arose by mutation and were shuffled by meiosis and random fertilisation. A fish cannot decide to change its alleles; the environment simply allows those with advantageous features to survive, reproduce and pass those alleles on.

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